Video 12 – Regularization

Video Content

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Video Synopsis

Introduction **_04:40 -14:00_**

Regularization is introduced as a method to 'put the brakes' on a learning algorithm in order to make sure the model does not capture noise present in the data. This is exemplified with an example that shows that regularization generally results in slightly higher bias, but significantly lower variance. The example shows the influence of regularization can be dramatic!

<u>SAFE SKIP!</u> Mathematical derivation of regularization as constrained and unconstrained optimisation problem **_14:00 – 32:21_**

It is shown that regularization can be formulated as a constraint optimisation problem. Graphically it is then shown that this constrained optimisation problem can be formulated as an unconstrained optimisation problem of minimising Ein plus an extra term consisting of the weights with a factor of proportionality.

Regularization applied to linear regression **_32:21 – 34:36_**

This section reiterates that regularization applied to linear regression means that we minimise the in-sample-error (as in normal linear regression) PLUS a term added by the regulariser. This term is dependent on the weights and minimising this term means that we will constrain the weights in some way. This new error will be called Augmented Error and it is shown that regularization is a way of reducing the VC dimension of the model.

Linear regression with regularization – the maths **_34:36 – 37:26_**

This section presents some maths showing how to solve the linear regression error equation with the regularization term added for the weights. Being able to follow the maths is not necessary, but it is useful to see that the new solution is very close to the old pseudo-inverse!

What is the effect of varying λ (lambda)? **_37:26 – 40:41_**

This section provides a pictorial representation of the influence of varying λ and is very useful for obtaining a conceptual understanding of regularization and its importance.

A number of regularizers **_40:41 – 52:10_**

The regularizer thus far considered is introduced as 'weight decay' and the name is explained using gradient descent as an example. Further regularizers are then introduced: Tikhonov is introduced as a general method to implement various regularizers, weight growth is introduced as a bad example of a regularizer, but to illustrate that regularizers generally favour smooth models such that the generally high-frequency noise is simply ignored.

Relationship between Augmented error and VC bound ***52:10 – 55:50***

An interesting parallel between the Augmented Error and the VC bound is shown and it is shown that the Augmented error is a proxy for Eout.

The perfect regularizer ***55:50-1:00:20***

Regularization is a general attempt without knowledge about the target function to reduce the effect of noise on overfitting. In doing so, the regularizer will also harm the ability of the model to capture the target function, but this will be less (if a proper regularizer was chosen) than the benefit the regularizer brings in terms of ignoring the noise in the data. Because noise tends to be a high-frequency signal, regularizers tend to favour smoother or simpler models. The question is raised what will happen when you choose a bad regularizer. The answer is validation (next video) which will allow us to choose the best model. In the case of a bad regularizer this may then simply be the model that does not make use of the regularizer at all.

Regularization in neural nets ***1:00:20 – 1:06:29***

Weight decay, weight elimination, soft weight elimination and early stopping are introduced as regularizers for neural networks.

Weight decay, which tries to keep the weights small, results in a neural network favouring small weights. Small weights means that the activation function (in case of a sigmoid) is in its linear region, which means that the model overall is relatively simple. Hence weight decay favours simple models.

In Weigth elimination, some of the weights are forces to be zero. This results in fewer free parameters, which in turn results in a lower VC dimension as the VC dimension is related to the number of free parameters in a neural net.

Soft weight elimination is introduced as a combination of weight decay and weight elimination. It only applies weight elimination to the small weights, which are pushed towards zero. The large weights are left alone.

Finally, early stopping is introduced as a regularizer. In early stopping, you observe (a proxy of) Eout as you train the network. Initially Eout will go down together with the ever decreasing Ein. At some moment Eout will start to go up and this is where you stop training and you declare you have your final hypothesis as further training will result in an increase in Eout.

How does λ relate to stochastic and deterministic noise? ***1:06:29 – 1:08:56***

Short section to show that more noise requires a higher lambda (more regularization). This section shows that the exact same holds for deterministic noise, i.e. target complexity.

<u>SAFE SKIP!</u> Q&A ***1:08:56 – end***